

RALEIGH EXECUTIVE, NC, USA

WMO: 722201

Lat: 35.582N

Lon: 79.101W

Elev: 247

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03723

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.2	21.4	1.1	5.9	27.9	6.9	7.9	30.6	20.0	53.3	17.9	52.2	1.4	320	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	96.2	74.9	93.1	74.5	90.9	74.1	78.9	88.4	77.8	87.0	76.9	85.8	6.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	140.8	81.9	75.3	133.7	80.8	74.5	130.3	80.3	42.4	88.7	41.2	87.4	40.4	84.9	97.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
16.7	14.0	11.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			DB	10.1	99.1	5.8	3.1	6.0	101.3	2.6	103.1	-0.6	104.9	-4.8	107.1	
			WB	11.2	81.2	6.5	1.1	6.5	82.0	2.7	82.6	-1.0	83.2	-5.8	84.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	41.1	43.9	50.6	60.1	68.6	76.4	79.6	78.3	72.7	60.8	49.5	44.4
	DBStd	15.78	10.16	9.34	9.70	8.12	6.51	4.93	4.04	4.64	5.62	8.53	8.34	9.12
	HDD50	1005	309	218	117	13	0	0	0	0	0	12	110	226
	HDD65	3320	740	594	455	188	43	1	0	0	9	184	467	639
	CDD50	4868	34	46	134	316	576	792	916	877	681	348	95	53
	CDD65	1711	0	2	8	41	155	343	451	412	240	55	3	1
	CDH74	16780	2	12	119	535	1495	3504	4611	3914	2067	488	28	5
	CDH80	7067	0	0	12	130	504	1566	2179	1787	775	113	1	0
Wind	WSAvg	3.5	3.9	4.4	4.7	4.7	3.6	3.1	2.8	2.4	3.1	3.2	3.3	3.3
Precipitation	PrecAvg	46.5	3.6	3.1	4.2	3.2	3.3	4.3	5.2	4.8	4.5	3.7	3.1	3.1
	PrecMax	59.5	7.8	6.3	7.6	6.5	7.4	12.4	8.5	10.7	16.5	9.3	8.0	6.3
	PrecMin	37.5	0.9	1.0	1.1	1.2	1.5	0.9	3.0	1.1	0.8	0.1	0.5	1.1
	PrecStd	6.1	1.9	1.3	1.8	1.4	1.4	2.7	1.5	2.1	3.7	2.4	2.0	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.1	75.3	81.5	88.0	91.6	98.2	99.2	99.8	94.9	88.3	78.7	73.4
		MCWB	62.7	63.5	61.3	65.2	71.1	74.6	76.5	77.9	70.6	72.3	64.1	64.8
	2%	DB	67.9	70.4	77.3	83.7	88.9	93.6	95.4	95.2	90.8	82.5	73.1	69.9
		MCWB	60.6	59.2	61.4	64.9	69.9	73.9	75.9	75.8	71.3	68.8	61.7	63.4
	5%	DB	63.5	65.8	73.2	80.8	85.8	91.1	93.0	91.7	88.0	79.3	70.0	65.0
		MCWB	56.5	56.6	58.8	63.7	69.5	73.6	75.7	75.6	72.0	67.7	61.4	60.9
	10%	DB	59.0	61.4	68.5	76.5	82.1	89.1	90.6	89.5	84.3	75.4	65.7	61.2
		MCWB	52.2	52.6	57.8	62.5	68.3	72.9	75.2	75.2	71.5	67.2	59.0	55.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.8	65.7	66.2	71.5	74.9	79.1	80.4	81.1	78.2	75.4	69.2	68.3
		MCDB	69.1	72.9	74.8	77.3	83.6	88.7	90.6	90.4	83.7	80.5	72.7	70.8
	2%	WB	62.3	62.4	64.3	69.0	73.0	77.1	78.9	79.0	76.8	73.9	66.0	64.9
		MCDB	66.8	68.0	71.4	75.9	81.9	86.9	89.3	88.0	83.1	78.7	69.8	68.0
	5%	WB	58.5	59.3	62.7	67.0	71.7	76.1	77.8	77.9	75.4	72.0	63.2	61.8
		MCDB	62.6	64.5	69.1	74.8	80.9	86.0	87.9	86.7	82.2	77.2	67.2	64.5
	10%	WB	53.8	54.4	60.4	65.0	70.4	75.1	76.9	76.8	74.2	69.0	60.0	57.1
		MCDB	57.0	59.3	67.4	73.2	78.7	84.7	86.5	85.2	80.2	73.2	65.2	60.3
Mean Daily Temperature Range	5% DB	MDBR	21.8	22.9	24.9	25.9	23.1	22.2	21.2	20.0	20.9	23.7	25.1	21.8
		MCDBR	27.5	28.4	30.4	30.5	27.0	25.1	24.1	23.6	25.8	26.2	28.7	25.4
		MCWBR	19.9	18.0	15.6	13.5	11.0	8.7	8.2	8.4	10.0	14.0	18.6	19.0
	5% WB	MCDBR	24.1	25.0	24.2	23.2	22.1	22.0	21.0	20.3	18.1	19.1	22.3	22.2
		MCWBR	20.0	18.5	15.0	12.4	10.1	8.7	8.2	8.5	8.9	11.4	17.6	18.6
Clear-Sky Solar Irradiance	taub	0.319	0.336	0.370	0.411	0.463	0.506	0.553	0.533	0.432	0.391	0.338	0.320	
	taud	2.484	2.431	2.370	2.256	2.149	2.070	1.958	1.999	2.279	2.362	2.474	2.516	
	Ebn at Noon	279	287	285	277	263	251	238	240	261	263	270	272	
	Edh at Noon	27	31	36	43	49	52	58	55	39	33	27	24	
All-Sky Solar Radiation	RadAvg	823	1036	1369	1721	1903	2027	1913	1734	1455	1182	904	712	
	RadStd	54	124	97	123	149	144	118	86	125	156	72	60	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (64 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+281	+200

Nomenclature: See separate page